

## AL NABK, Syrian Arab Republic

WMO: 400830

Lat: 34.033N

Lon: 36.717E

Elev: 1333

StdP: 86.30

Time Zone: 2.00 (SYR)

Period: 94-12

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB  |             | Humidification DP/MCDB and HR |            |            |              |            |            | Coldest Month WS/MCDB |            |            |            | MCWS/PCWD to 99.6% DB |            | WSF          |
|---------------|-------------|-------------|-------------------------------|------------|------------|--------------|------------|------------|-----------------------|------------|------------|------------|-----------------------|------------|--------------|
|               |             |             | 99.6%                         |            |            | 99%          |            |            | 0.4%                  |            | 1%         |            |                       |            |              |
|               | 99.6%       | 99%         | DP                            | HR         | MCDB       | DP           | HR         | MCDB       | WS                    | MCDB       | WS         | MCDB       | MCWS                  | PCWD       |              |
| (a)<br>1      | (b)<br>-4.0 | (c)<br>-2.3 | (d)<br>-12.8                  | (e)<br>1.5 | (f)<br>6.9 | (g)<br>-10.4 | (h)<br>1.8 | (i)<br>6.5 | (j)<br>10.6           | (k)<br>3.7 | (l)<br>9.3 | (m)<br>5.1 | (n)<br>2.1            | (o)<br>180 | (p)<br>0.397 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 12.7                   | 32.8            | 16.2 | 31.4 | 15.8 | 30.2 | 15.6 | 18.8                | 26.2 | 18.0 | 26.1 | 17.3 | 26.0 | 2.7                  | 270  |

| Dehumidification DP/MCDB and HR |      |      |      |      |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|------|------|------|------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |      |      | 1%   |      |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR   | MCDB | DP   | HR   | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)  | (c)  | (d)  | (e)  | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  | (p)               |
| 16.5                            | 13.8 | 22.0 | 15.3 | 12.8 | 21.8 | 14.2 | 11.9 | 21.3 | 59.3          | 25.8 | 56.5 | 26.1 | 54.1 | 25.8 | 22.6              |

## Extreme Annual Design Conditions

| Extreme Annual WS |         |         | Extreme Annual Temperature |          |                    |         | n-Year Return Period Values of Extreme Temperature |          |            |          |            |          |            |          |    |    |
|-------------------|---------|---------|----------------------------|----------|--------------------|---------|----------------------------------------------------|----------|------------|----------|------------|----------|------------|----------|----|----|
|                   |         |         | Mean                       |          | Standard Deviation |         | n=5 years                                          |          | n=10 years |          | n=20 years |          | n=50 years |          |    |    |
| 1%                | 2.5%    | 5%      | Min                        | Max      | Min                | Max     | Min                                                | Max      | Min        | Max      | Min        | Max      | Min        | Max      | DB | WB |
| (a) 8.3           | (b) 6.6 | (c) 5.4 | (e) N/A                    | (f) N/A  | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A  | (k) N/A    | (l) N/A  | (m) N/A    | (n) N/A  | (o) N/A    | (p) N/A  |    |    |
| (a) 8.3           | (b) 6.6 | (c) 5.4 | (e) -7.2                   | (f) 21.1 | (g) 2.1            | (h) 1.0 | (i) -8.7                                           | (j) 21.8 | (k) -9.9   | (l) 22.4 | (m) -11.1  | (n) 22.9 | (o) -12.6  | (p) 23.6 |    |    |

## Monthly Climatic Design Conditions

|                                                                                  |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |
|----------------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|
|                                                                                  |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |
| Temperatures,<br>Degree-Days<br>and<br>Degree-Hours                              | DBAvg       | 14.3   | 4.1   | 5.4   | 8.4   | 13.0  | 17.0  | 21.2  | 23.7  | 23.9  | 20.6  | 16.5  | 10.2  | 6.5  |
|                                                                                  | DBStd       | 7.57   | 3.15  | 3.25  | 3.82  | 3.60  | 3.16  | 2.57  | 2.11  | 2.25  | 2.55  | 3.07  | 3.32  | 3.11 |
|                                                                                  | HDD10.0     | 559    | 185   | 134   | 77    | 10    | 1     | 0     | 0     | 0     | 0     | 1     | 37    | 114  |
|                                                                                  | HDD18.3     | 2039   | 442   | 363   | 308   | 166   | 63    | 5     | 0     | 0     | 8     | 73    | 246   | 366  |
|                                                                                  | CDD10.0     | 2114   | 2     | 5     | 28    | 99    | 218   | 337   | 426   | 429   | 318   | 203   | 42    | 7    |
|                                                                                  | CDD18.3     | 551    | 0     | 0     | 1     | 4     | 22    | 92    | 168   | 171   | 76    | 17    | 0     | 0    |
|                                                                                  | CDH23.3     | 5215   | 0     | 0     | 7     | 39    | 202   | 876   | 1683  | 1671  | 623   | 113   | 1     | 0    |
|                                                                                  | CDH26.7     | 1694   | 0     | 0     | 0     | 6     | 27    | 232   | 624   | 652   | 146   | 8     | 0     | 0    |
| Wind                                                                             | WSAvg       | 1.9    | 2.1   | 2.1   | 2.3   | 2.4   | 2.0   | 2.0   | 2.0   | 1.5   | 1.7   | 1.5   | 1.7   | 1.6  |
| Precipitation                                                                    | PrecAvg     | 255    | 48    | 44    | 35    | 17    | 13    | 2     | 3     | 1     | 2     | 18    | 32    | 37   |
|                                                                                  | PrecMax     | 410    | 84    | 110   | 111   | 59    | 65    | 18    | 55    | 13    | 15    | 70    | 104   | 88   |
|                                                                                  | PrecMin     | 143    | 17    | 13    | 11    | 3     | 2     | 0     | 0     | 0     | 0     | 1     | 0     | 11   |
|                                                                                  | PrecStd     | 68     | 19    | 25    | 23    | 12    | 17    | 4     | 11    | 3     | 4     | 17    | 23    | 21   |
| Monthly Design<br>Dry Bulb<br>and<br>Mean Coincident<br>Wet Bulb<br>Temperatures | 0.4%        | DB     | 15.5  | 18.3  | 24.1  | 27.4  | 29.1  | 32.2  | 35.2  | 34.7  | 32.1  | 27.6  | 22.6  | 19.3 |
|                                                                                  |             | MCWB   | 6.6   | 9.6   | 10.5  | 11.7  | 13.6  | 15.1  | 16.8  | 16.4  | 15.7  | 13.2  | 10.4  | 7.9  |
|                                                                                  | 2%          | DB     | 12.5  | 14.3  | 20.5  | 23.8  | 27.1  | 30.5  | 32.9  | 32.9  | 29.8  | 26.2  | 19.8  | 15.7 |
|                                                                                  |             | MCWB   | 5.9   | 6.6   | 9.2   | 10.8  | 12.8  | 14.7  | 16.3  | 16.4  | 15.1  | 13.1  | 9.8   | 7.1  |
|                                                                                  | 5%          | DB     | 10.5  | 12.4  | 17.7  | 21.9  | 25.7  | 29.2  | 31.4  | 31.4  | 28.2  | 24.4  | 17.7  | 13.2 |
|                                                                                  |             | MCWB   | 5.1   | 5.8   | 8.1   | 10.3  | 12.5  | 14.6  | 16.1  | 15.9  | 14.7  | 12.8  | 9.3   | 6.5  |
|                                                                                  | 10%         | DB     | 8.8   | 10.5  | 15.4  | 20.0  | 24.0  | 28.0  | 30.1  | 30.1  | 27.0  | 22.7  | 15.9  | 11.2 |
|                                                                                  |             | MCWB   | 4.5   | 5.1   | 7.2   | 9.7   | 12.2  | 14.4  | 15.8  | 15.9  | 14.3  | 12.5  | 8.8   | 6.0  |
| Monthly Design<br>Wet Bulb<br>and<br>Mean Coincident<br>Dry Bulb<br>Temperatures | 0.4%        | WB     | 8.0   | 9.4   | 11.5  | 13.3  | 15.8  | 17.8  | 20.1  | 20.1  | 18.4  | 15.6  | 12.3  | 9.6  |
|                                                                                  |             | MCDB   | 12.5  | 16.0  | 20.6  | 21.3  | 23.4  | 25.3  | 28.4  | 27.7  | 24.8  | 21.7  | 17.7  | 15.2 |
|                                                                                  | 2%          | WB     | 6.7   | 7.6   | 10.0  | 12.2  | 14.6  | 16.7  | 18.8  | 19.0  | 17.5  | 14.7  | 11.2  | 8.4  |
|                                                                                  |             | MCDB   | 10.9  | 13.1  | 17.5  | 20.6  | 21.9  | 24.9  | 26.6  | 25.9  | 24.2  | 21.3  | 16.5  | 12.9 |
|                                                                                  | 5%          | WB     | 5.7   | 6.5   | 9.0   | 11.3  | 13.9  | 16.0  | 18.0  | 18.1  | 16.7  | 14.0  | 10.2  | 7.5  |
|                                                                                  |             | MCDB   | 9.5   | 11.0  | 16.3  | 19.7  | 21.9  | 25.4  | 26.7  | 26.1  | 23.9  | 20.9  | 15.8  | 11.7 |
|                                                                                  | 10%         | WB     | 5.0   | 5.5   | 8.0   | 10.5  | 13.2  | 15.3  | 17.2  | 17.4  | 15.9  | 13.4  | 9.4   | 6.7  |
|                                                                                  |             | MCDB   | 8.1   | 9.6   | 14.2  | 18.3  | 21.4  | 25.0  | 26.9  | 26.2  | 24.0  | 20.5  | 14.8  | 10.4 |
| Mean Daily<br>Temperature<br>Range                                               | 5% DB       | MDBR   | 7.8   | 8.3   | 9.5   | 10.8  | 11.9  | 12.8  | 12.7  | 13.0  | 12.4  | 10.9  | 9.6   | 8.7  |
|                                                                                  |             | MCDBR  | 10.0  | 10.1  | 11.5  | 12.4  | 13.1  | 13.4  | 13.5  | 13.5  | 13.4  | 12.4  | 11.7  | 10.9 |
|                                                                                  |             | MCWBR  | 5.9   | 5.8   | 5.5   | 5.6   | 5.6   | 6.0   | 6.3   | 6.1   | 6.3   | 5.6   | 6.1   | 6.4  |
|                                                                                  | 5% WB       | MCDBR  | 8.8   | 9.2   | 10.7  | 11.7  | 11.9  | 13.1  | 12.8  | 13.0  | 12.6  | 11.2  | 10.1  | 9.1  |
|                                                                                  |             | MCWBR  | 5.6   | 6.0   | 6.1   | 5.7   | 6.0   | 6.6   | 7.3   | 7.3   | 6.9   | 5.9   | 5.9   | 6.0  |
| Clear-Sky<br>Solar<br>Irradiance                                                 | taub        | 0.300  | 0.323 | 0.363 | 0.409 | 0.407 | 0.353 | 0.354 | 0.361 | 0.352 | 0.369 | 0.324 | 0.307 |      |
|                                                                                  | taud        | 2.354  | 2.278 | 2.148 | 2.016 | 2.063 | 2.245 | 2.249 | 2.224 | 2.253 | 2.194 | 2.315 | 2.347 |      |
|                                                                                  | Ebn at Noon | 909    | 920   | 906   | 877   | 880   | 925   | 923   | 912   | 907   | 854   | 869   | 875   |      |
|                                                                                  | Edh at Noon | 96     | 116   | 144   | 173   | 167   | 139   | 138   | 138   | 128   | 125   | 100   | 92    |      |
| All-Sky Solar<br>Radiation                                                       | RadAvg      | 2.65   | 3.50  | 4.99  | 6.17  | 7.31  | 8.29  | 8.10  | 7.35  | 6.17  | 4.53  | 3.25  | 2.47  |      |
|                                                                                  | RadStd      | 0.18   | 0.30  | 0.28  | 0.32  | 0.40  | 0.14  | 0.11  | 0.17  | 0.21  | 0.22  | 0.22  | 0.18  |      |

## Historical Trends

|                              | DBAvg | Heating |       |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|-------|---------|-------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |       | 99% DB  |       | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
|                              |       | (d)     | (e)   | (f)    | (g)     | (h)   | (i)   | (j)         | (k)     | (l)     | (m)     |
| (46) Station Only            | N/A   | N/A     | N/A   | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (13 neighbors) | +0.30 | N/A     | -1.33 | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | +173    | +110    |

Nomenclature: See separate page